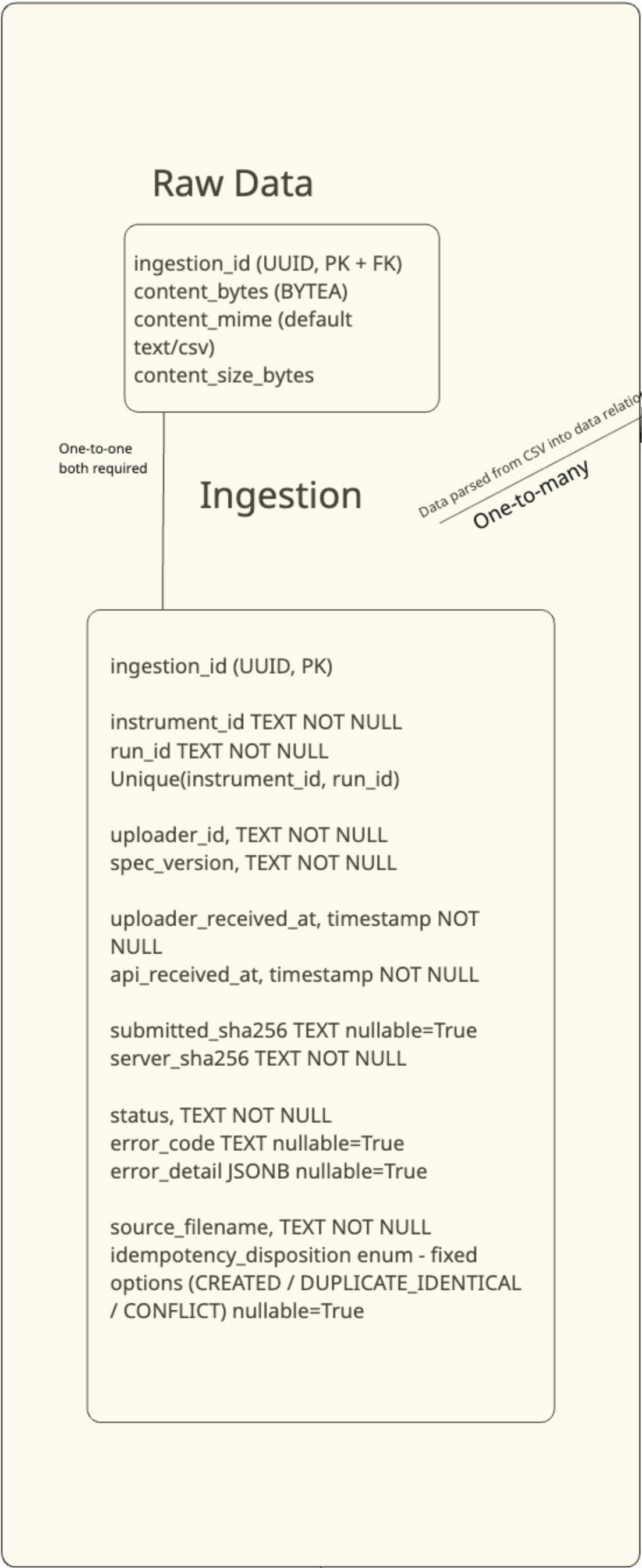
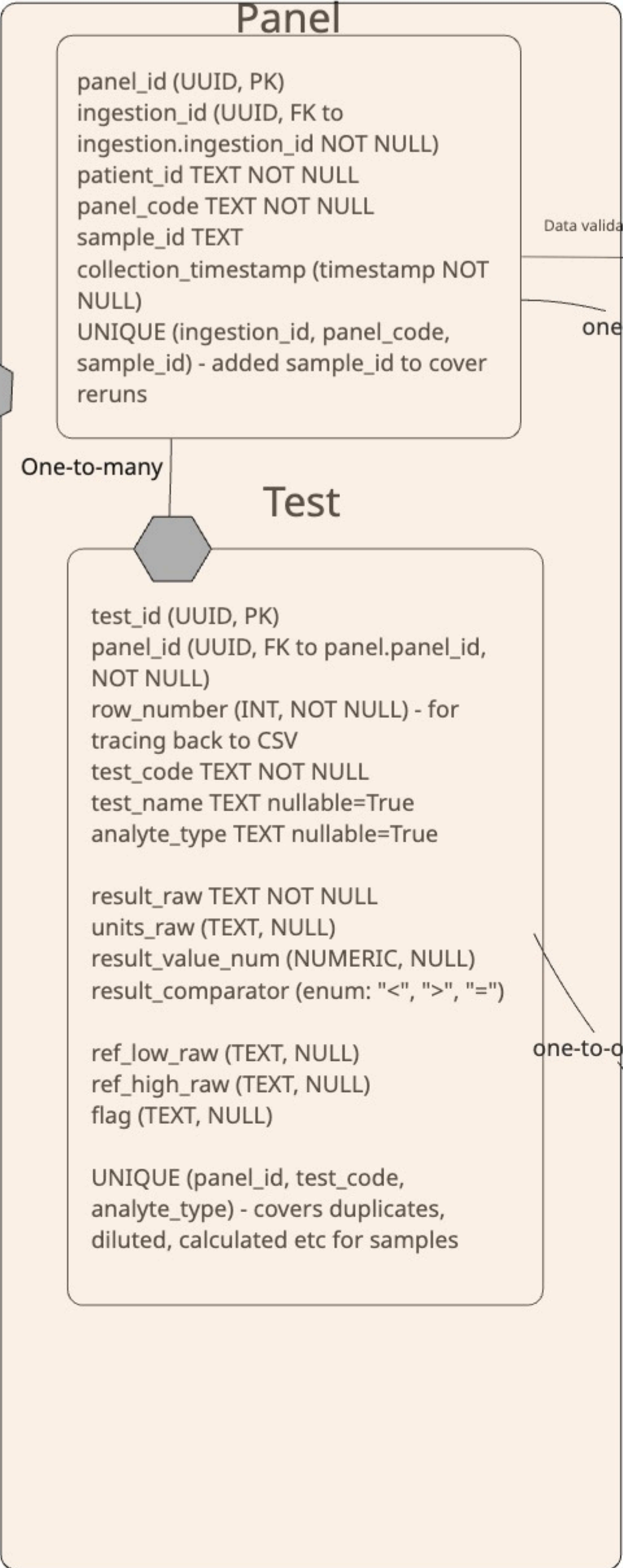


Data pipeline: raw ingest - parsed relations - validated and normalized FHIR artifacts

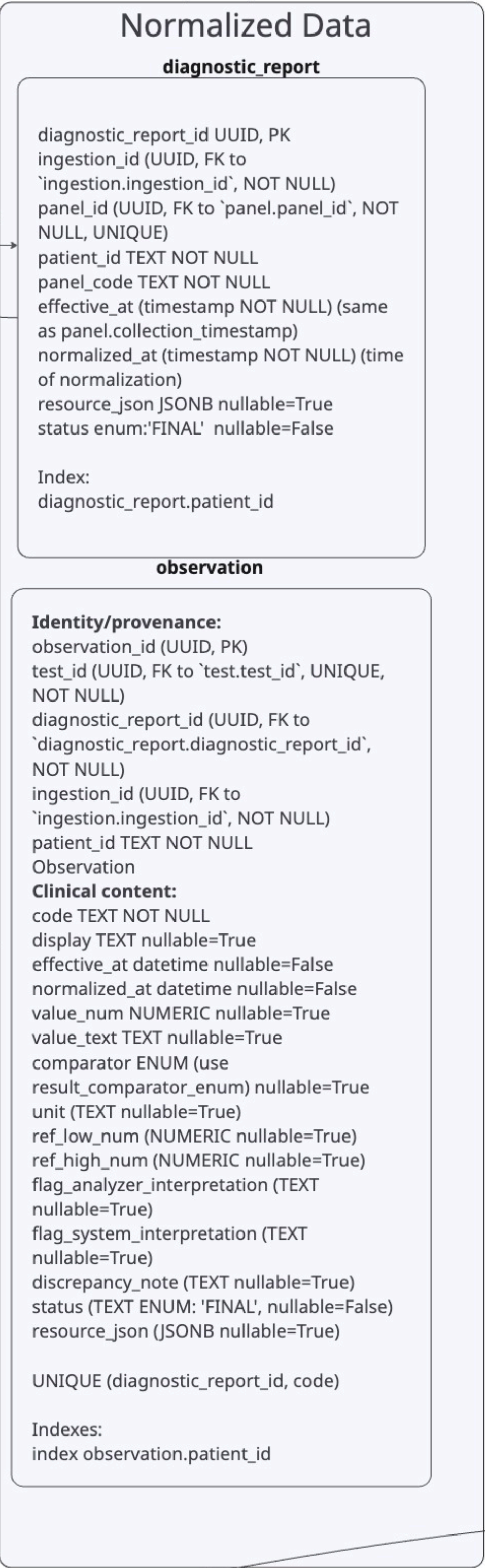
Immutable: source of truth



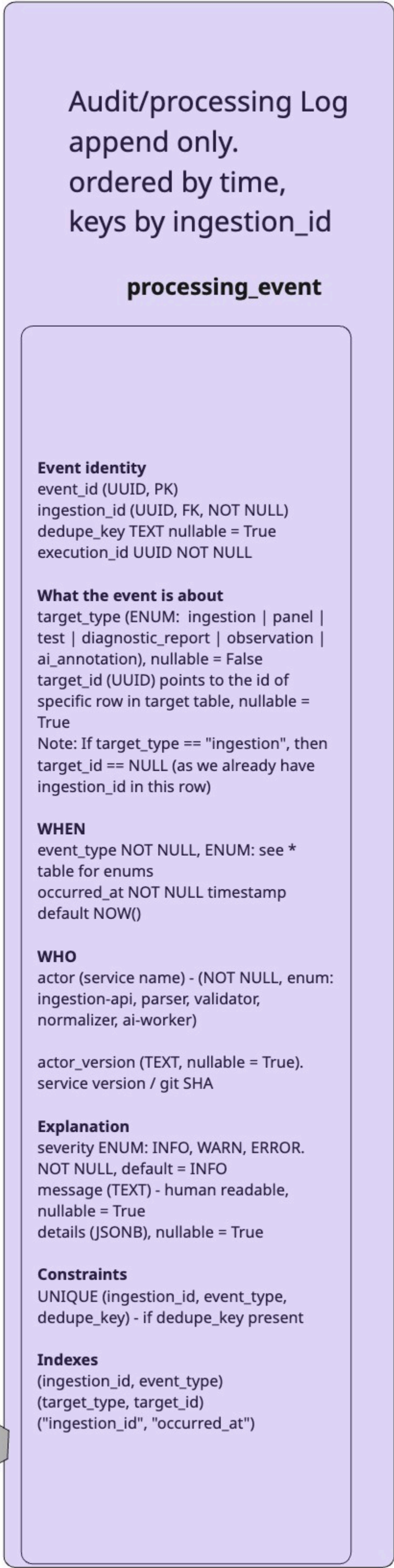
Parsed/staging representation of data. Structured but not normalized.



Normalized clinical artifacts. FHIR-shaped persistence

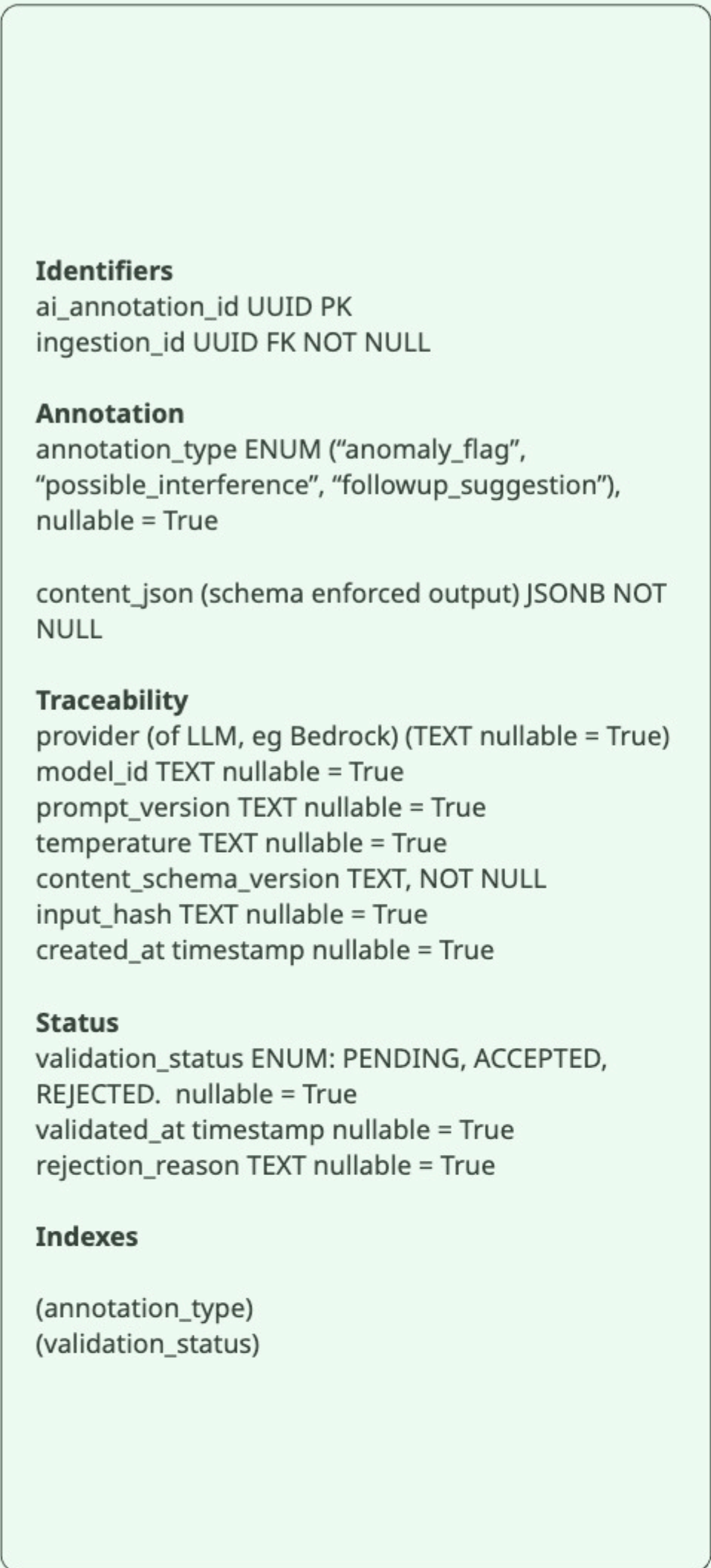


Append-only operational history



AI augmentation artifacts

AI annotation

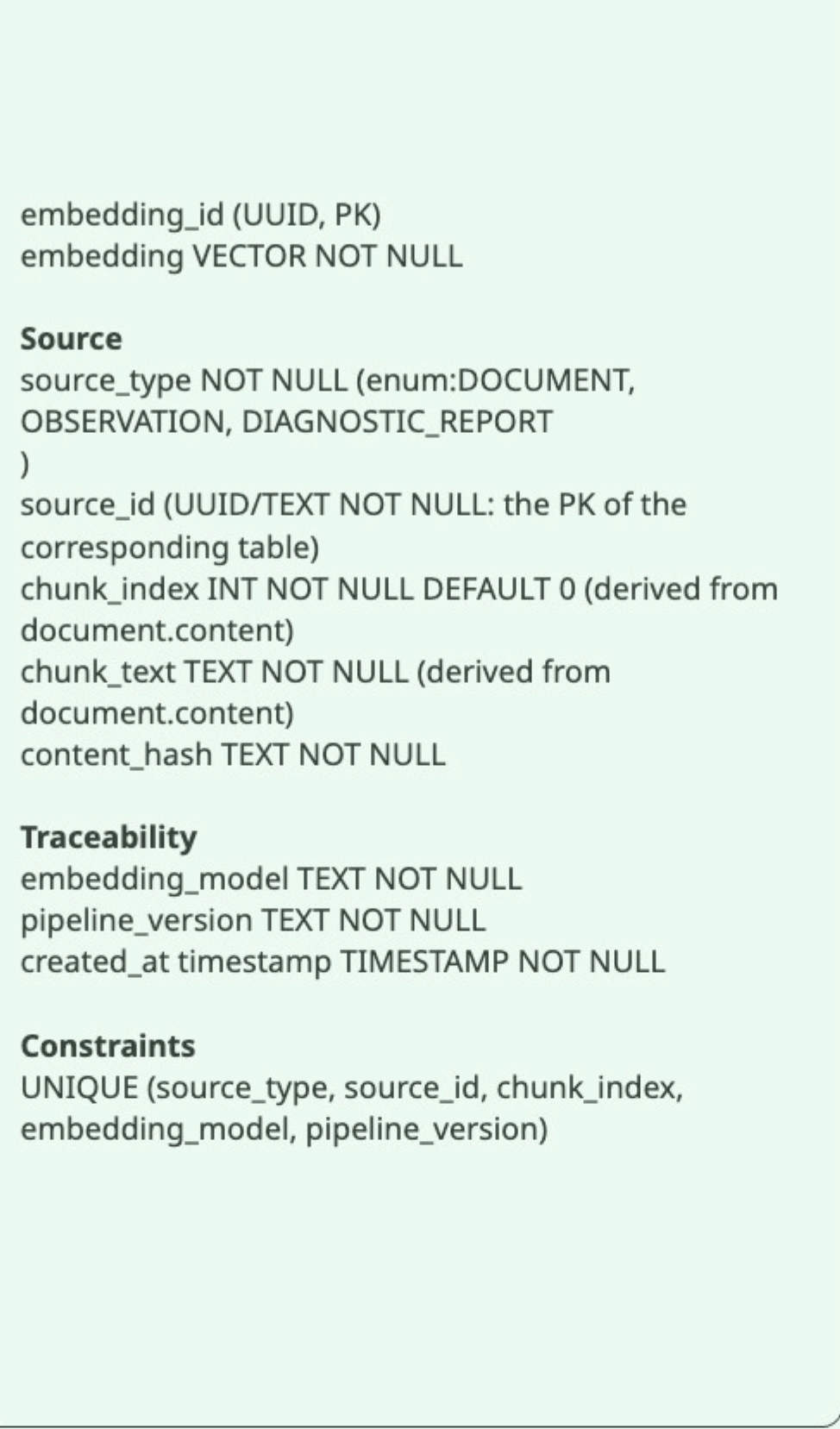


Ai generation job

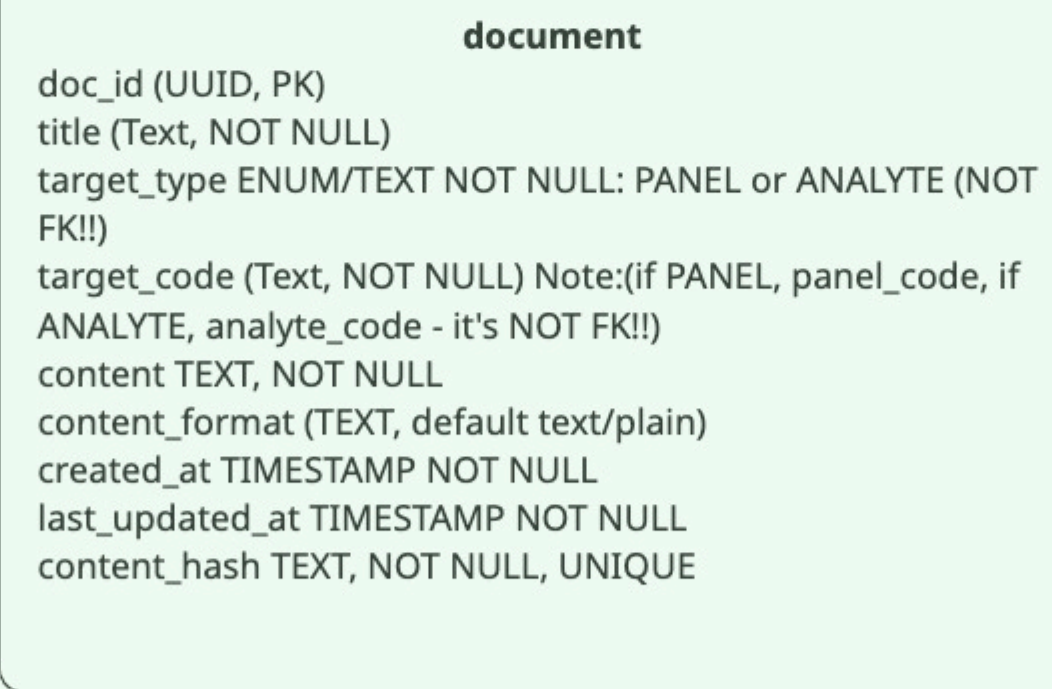
correlation_id uuid

Stores embedded vectors for reference ranges, clinical interpretation guidelines, historical normalized lab results

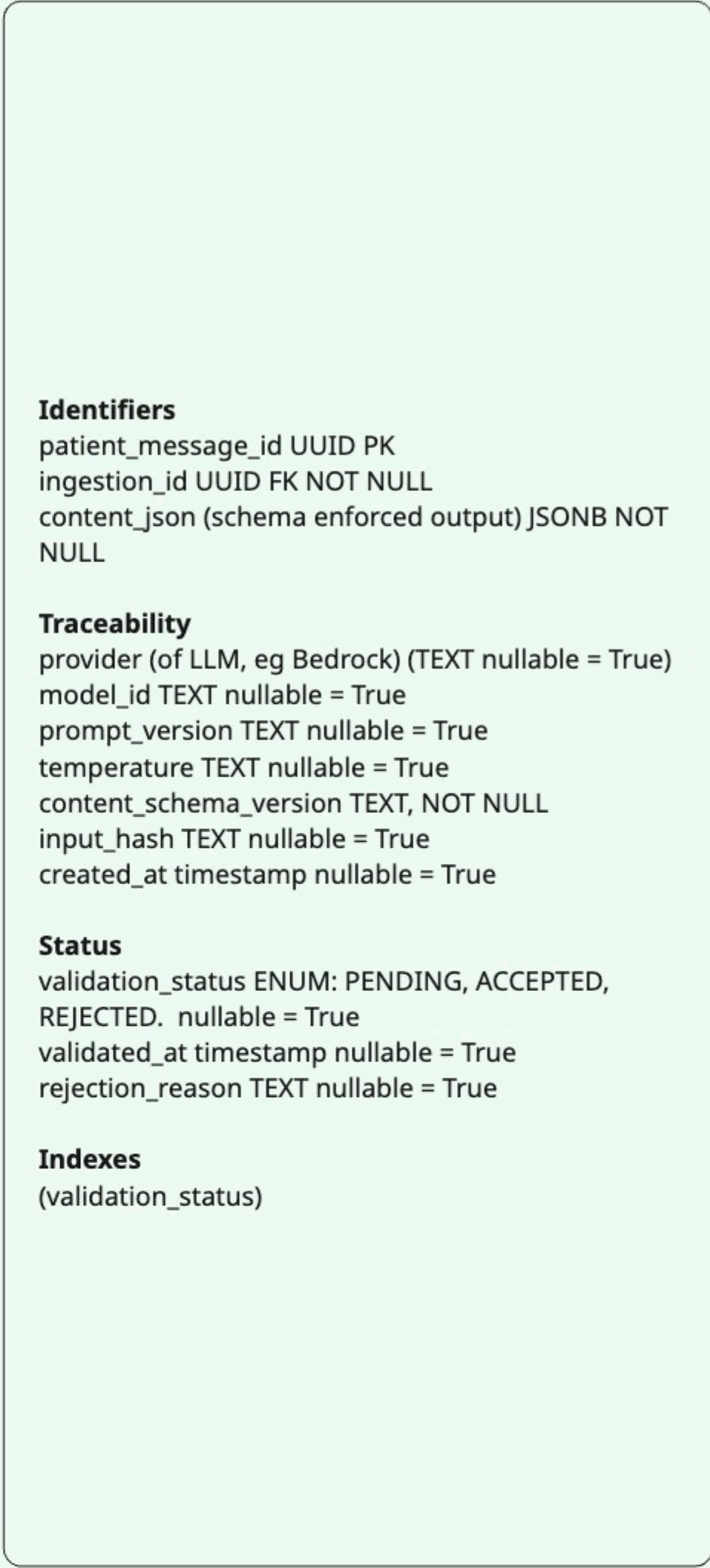
Vector Store (embeddings + metadata)



Documents for vector embeddings



Patient Message



Patient (minimal)

patient_id
first_name
last_name
email

One-to-many(0)

*Event types for processing_event table:

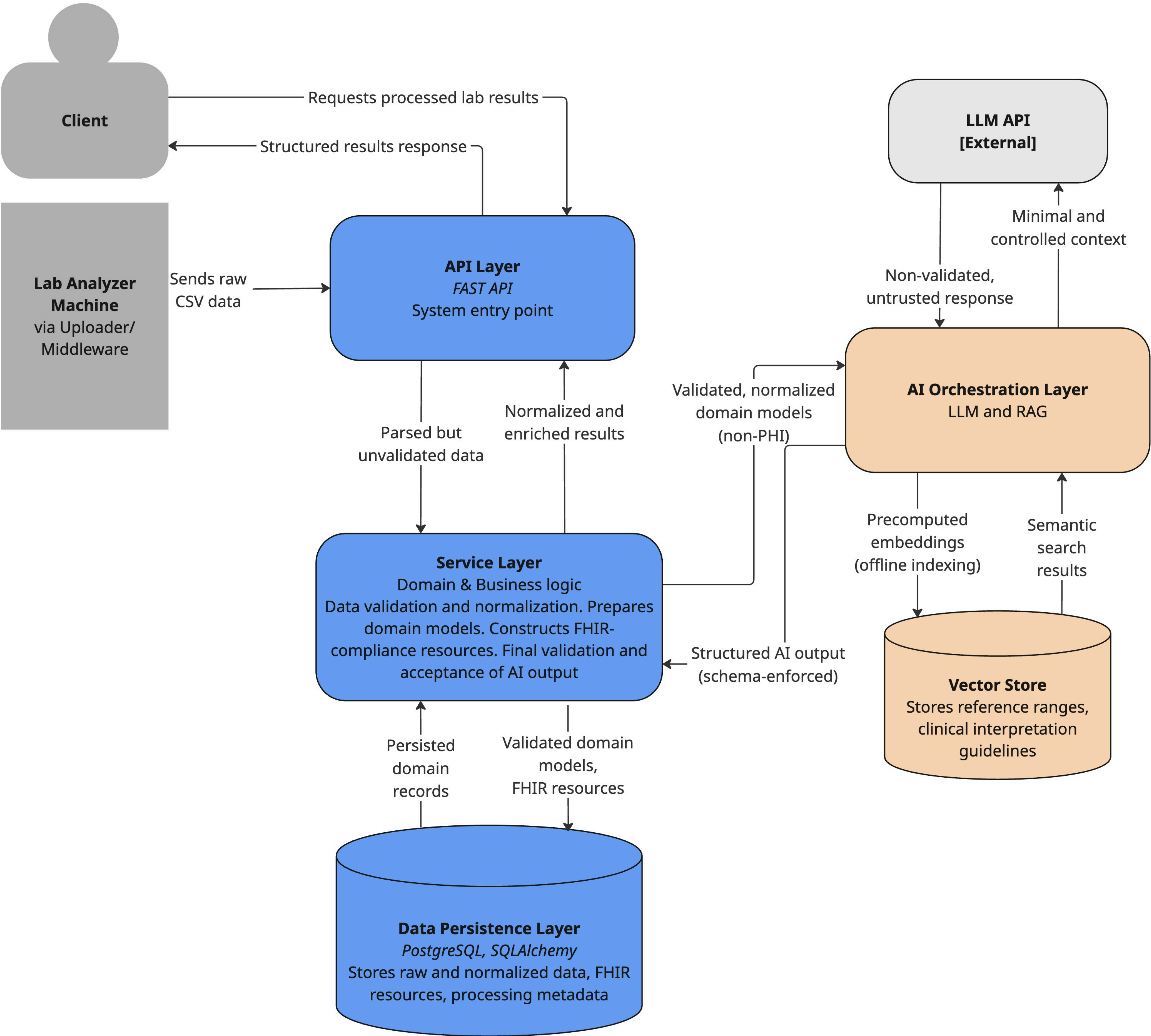
Ingestion
INGESTION_ACCEPTED
INGESTION_DEDUPED_IDENTICAL
INGESTION_CONFLICT

Parse
PARSE_STARTED
PARSE_SUCCEEDED (include row counts)
PARSE_FAILED (include error summary)

Validation
VALIDATION_STARTED
VALIDATION_SUCCEEDED
VALIDATION_FAILED (include counts + top error codes)

Normalization
NORMALIZATION_STARTED
NORMALIZATION_SUCCEEDED
NORMALIZATION_FAILED

AI augmentation
AI_ENRICHMENT_STARTED
AI_ENRICHMENT_SKIPPED (reason)
AI_ENRICHMENT_SUCCEEDED
AI_ENRICHMENT_FAILED



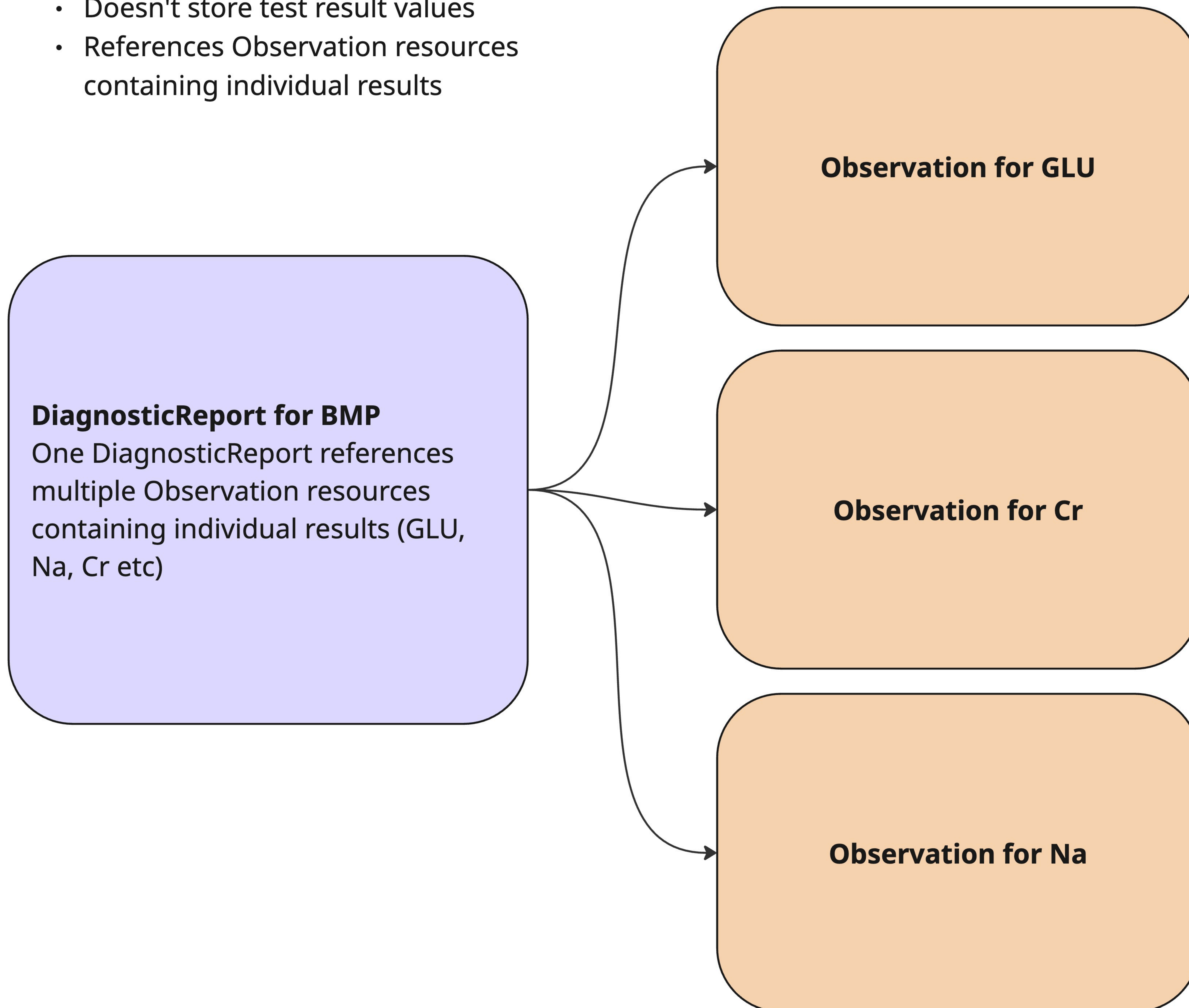
FHIR Resources

DiagnosticReport

- One per test panel
- Groups Observation resources and provides clinical context
- Doesn't store test result values
- References Observation resources containing individual results

Observation Resource

- Contains individual test result



Note: Chronologically, Observations are created first, then grouped under a DiagnosticReport via references.